Sushil Kumar

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Skills

Languages: C/C++, C#, Python
Databases: MySQL, SQL, Redis
Platforms: Android, Linux
Concepts: OOP, Data Structure & Algorithms, System Design, Operating System, Multithreading, IPC, Microservices, REST APIs, DBMS, Machine Learning, Deep Learning, LLMs, Generative AI, PyTorch
DevOps/Tools: Kafka, RabbitMQ, Git, GitHub, CI/CD, AWS, Jira, TeraTerm, Visual Studio, Confluence
Soft Skills: Leadership, Innovation, Teamwork, Effective Communication, Root-Cause Analysis, Data-Driven Decision-Making

Professional Summary: Software Engineer with ∼**4 years of experience** in full-cycle development, system debugging, and rapid POC delivery. Experienced in leading teams, mentoring interns, and collaborating cross-functionally to build scalable, high-impact software solutions.

Experience

Samsung R&D Institute, Delhi - Software Engineer July 2022 - Present

- **SmartMirror App (CES 2025 Best of Innovation Award):** Engineered an AI-powered MICRO LED smart mirror—the world's first mirror-display hybrid—enabling real-time user detection (within 1 meter), advanced skin analysis, and personalized treatment recommendations.
- Implemented computer vision pipelines using object detection (DeviceCare, sub-70ms), image segmentation (sunscreen analysis, sub-150ms), and MediaPipe for face detection and tracking (sub-50ms), achieving real-time performance on-device.
- Integrated camera streaming, voice interaction, and LLM-based capabilities to deliver personalized insights on health (blood pressure, mental wellness), daily schedule, and product recommendations. Built on Linux and Android platforms. *[C++, C#, Python]*
- **POCs for Skin Health:** Led development of on-device **DeepLabV3+** with MobileNetV3 backbone for multi-class segmentation of acne, wrinkles, and pores, enabling real-time pixel-level visual overlays with sub-120ms inference latency.
- Trained using a combination of public dermatology datasets (Dermnet, ACNE04) and custom annotated data, achieving 70% mean IoU on a held-out test set and optimized inference latency using INT8 quantization. *[Python, OpenCV]*
- **The Player App:** Developed an intelligent portable media player (POC) for CES that detects nearby users and estimates relative distance to optimize multimedia experiences. Designed real-time master-client synchronization using WebSockets with timestamp alignment and drift correction, ensuring low-latency, near-synchronous playback across multiple devices. Integrated LLM capabilities to enhance interactive media experiences. *[C++, C#]*
- **360 Degree audio support in Samsung Galaxy earbuds:** Contributed to the development of 360 degree immersive audio support with head tracking for Samsung Galaxy earbuds, enhancing the video and music experience. [C++, BLE]
- **Enhanced Audio and Visual experience in Samsung TV:** Enhanced audio-visual preferences for sports, movie, and general modes using MVC architecture, resulting in a 20% increase in user satisfaction.
- **SmartAV Daemon:**Re-architected legacy C++ modules into a C++ & C# hybrid system with a daemon service to manage AV experiences on Samsung TV. Leveraged SOLID principles and integrated 150+ APIs to boost performance by 20% and reduce bugs by 25%.Improved IPC via D-Bus, achieving a SAM score of 4.85. [C++, C#]

Publications

Resolving Semantic Confusions for Improved Zero-Shot Detection BMVC-2022
Published: Nov 24, 2022 *Paper link*

- Proposed a novel zero-shot object detection framework using Faster-RCNN trained on limited classes. Leveraged a Conditional WGAN to generate synthetic visual features from word vectors, addressing semantic confusion through triplet and cyclic-consistency loss. Achieved an 8% relative mAP gain over the next-best method for detecting unseen objects on MSCOCO.

Education

2020–2022: **M.Tech** (Computer Science), **IIT Guwahati** (CGPA: 8.71)
2015–2019: **B.Tech** (Computer Science), **IIIT Kalyani** (CGPA: 8.21)

Projects

Banking System GitHub

- Built a banking system using C++ and TCP socket programming, supporting 50+ concurrent clients by designing a multithreaded client-server architecture for efficient request processing and communication.

Achievements & Extracurricular Activities

- Secured **AIR 387** among 100K candidates who appeared in **GATE 2020** examination.
- Solved over **800 coding problems** on platforms: LeetCode, InterviewBit, GFG.
- Awarded Best Code Review in 2024 for writing high-quality, efficient, scalable, and clean code at SRID.
- **Spot Award (2023; twice in 2025)** at SRI-Delhi for exceptional contributions to project delivery and team impact.
- Completed NPTEL Online Certification in Algorithm and Analysis with a 97% score.
- **Teaching Assistant** for Computing Lab (CS 110), Machine Learning & Deep Learning (CS 590) course at IIT Guwahati.